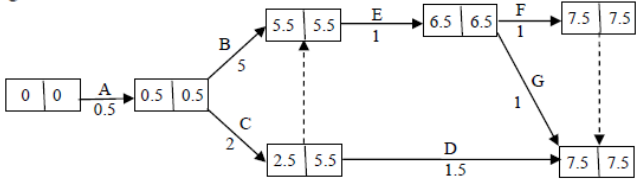
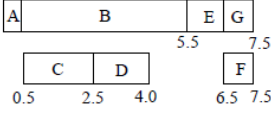


Question		Answer	Marks	Guidance
6	(i)	e.g.	M1 A1 A1 A1	activity-on-arc A, B, C D, E Rest
6	(ii)	e.g. minimum completion time – 12 days critical activities – A, C, D, F, H.	M1 A1 M1 A1 B1 B1	forward pass backward pass
6	(iii)	e.g. Minimum completion time = 10.5 days Afzal needs to be employed for 6.5 days.	B1 B1 B1 B1 B1	A, B, C D, E F, G, H
6	(iv)	17 days	B1	

5	(i)	<table><thead><tr><th>Activity</th><th>Duration (weeks)</th><th>Immediate predecessors</th></tr></thead><tbody><tr><td>A</td><td>1</td><td>—</td></tr><tr><td>B</td><td>1</td><td>—</td></tr><tr><td>C</td><td>1</td><td>—</td></tr><tr><td>D</td><td>8</td><td>A</td></tr><tr><td>E</td><td>6</td><td>B</td></tr><tr><td>F</td><td>6</td><td>C, I</td></tr><tr><td>G</td><td>6</td><td>—</td></tr><tr><td>H</td><td>1</td><td>D, E, F, G</td></tr><tr><td>I</td><td>2</td><td>—</td></tr><tr><td>J</td><td>4</td><td>A, B, C</td></tr><tr><td>K</td><td>2</td><td>H, J</td></tr></tbody></table>	Activity	Duration (weeks)	Immediate predecessors	A	1	—	B	1	—	C	1	—	D	8	A	E	6	B	F	6	C, I	G	6	—	H	1	D, E, F, G	I	2	—	J	4	A, B, C	K	2	H, J	B1 B3	A, B, C, G and I -1 for 1 or 2 wrong rows -2 for 3 or 4 wrong rows -3 for 5 or 6 wrong rows
Activity	Duration (weeks)	Immediate predecessors																																						
A	1	—																																						
B	1	—																																						
C	1	—																																						
D	8	A																																						
E	6	B																																						
F	6	C, I																																						
G	6	—																																						
H	1	D, E, F, G																																						
I	2	—																																						
J	4	A, B, C																																						
K	2	H, J																																						
5	(ii)+ (iii)	Critical activities ... A, D, H, K	M1 A1 A1 A1 A1 M1 A1✓ M1 A1✓ B1cao	activity on arc immediate predecessors A, B, C, I, G immediate predecessors D (A), E(B), F(C,I) immediate predecessors J(A,B,C), H(D,E,F,G) immediate predecessors K(J,H) + rest forward pass backward pass																																				

5	(iv)	eg													B1cao B1cao	F (ES2, LF9) rest
		Week	1	2	3	4	5	6	7	8	9	10	11	12		
		CPA start time	0	1	2	3	4	5	6	7	8	9	10	11		
		Room 1	A	D	D	D	D	D	D	D	D	H	K	K		
		Room 2	B	E	E	E	E	E	E	F	F					
		Room 3	C		F	F	F	F								
			G	G	G	G	G	G								
			I	I												
				J	J	J	J									

5	(i)&(ii)	minimum completion time = 55 minutes critical activities – A, E, F, G, H, J, K	M1 A1 A1 A1 A1 [5] M1A1✓ M1A1✓ B1cao B1cao [6]	activity on arc F & I J K rest forward pass backward pass time critical activities																																																
5	(iii)	e.g. (each cell represents 5 minutes) <table><tr><td>1st person</td><td>A</td><td>E</td><td>E</td><td>F</td><td>G</td><td></td><td></td><td></td><td></td><td>J</td><td>K</td></tr><tr><td>2nd person</td><td></td><td>B</td><td>D</td><td>I</td><td>I</td><td>I</td><td>I</td><td>I</td><td>I</td><td></td><td></td></tr><tr><td>other activities</td><td></td><td></td><td>C</td><td>C</td><td>C</td><td>C</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td>H</td><td>H</td><td>H</td><td>H</td><td></td><td></td></tr></table>	1 st person	A	E	E	F	G					J	K	2 nd person		B	D	I	I	I	I	I	I			other activities			C	C	C	C												H	H	H	H			M1 A1 B1 [3]	A, E, F, G allocated OK B, D, I, J, K OK C and H correctly timed
1 st person	A	E	E	F	G					J	K																																									
2 nd person		B	D	I	I	I	I	I	I																																											
other activities			C	C	C	C																																														
						H	H	H	H																																											
5	(iv)	e.g. <table><tr><td>1st person</td><td>A</td><td>D</td><td>I</td><td>I</td><td>I</td><td>I</td><td>I</td><td>I</td><td>J</td><td>K</td></tr><tr><td>2nd person</td><td></td><td>B</td><td>E</td><td>E</td><td>F</td><td>G</td><td></td><td></td><td></td><td></td></tr></table> 50 minutes	1 st person	A	D	I	I	I	I	I	I	J	K	2 nd person		B	E	E	F	G					B1 B1 [2]	a correct schedule for two people 50 minutes seen																										
1 st person	A	D	I	I	I	I	I	I	J	K																																										
2 nd person		B	E	E	F	G																																														

4	(i) & (ii)	e.g.  minimum completion time = 7.5 hours critical activities – A, B, E, F, G (or ABEG + ABEF)	M1 A1 A1 A1 A1 M1 A1✓ M1 A1✓ B1 B1	Activity on arc Single start and end A, B, C, D (precedences) E (precedences) F and G (all correct) forward pass backward pass time (cao) critical activities (cao)
4	(iii)	e.g.  Needs to be clear what is done by whom. This doesn't necessarily require people being labelled ... but might.	B1 B1	not fit must be labelled or to scale (e.g. on the squares provided) Can be written out instead.
4	(iv)	8.0 hours or delay 0.5 hours A, C, D 8.5 hours or delay of 1 hour	B1 B1 B1	cao ISW if needed cao cao ISW if needed